

be used with a stylus. To actually use a thumb in daily life, the screen had to be more robust. The capacitive touchscreen technology solved this problem as it was more robust, ushering in the era of mobile devices with capacitive screens that offer multi-touch capabilities and can be operated with the thumb.

Display trends will dictate the touch-trends. "Because touch technology follows display trends, the current display market serves as a good indicator for what to expect in touch," believes Sewell. As demand for collaborative and intuitive display surfaces is on the rise, touchscreen technologies that enable multi-touch are likely to gain more traction.

Projected-capacitive will lead the pack. "Projected-capacitive has been the technology of choice since 2010. It will continue to dominate the consumer market and become the standard in the vertical sectors within the next five years," says Sewell.

Resistive technologies will slowly phase out. Resistive-based touchscreens will gradually be phased out or used less, primarily because of their robustness issues. "Robust technologies that have a glass on top, like through-glass capacitance technologies, will become more popular," says Tapadia.

Interesting application areas will open up. "Till now touchscreen was being used for personal information data consumption in devices such as mobile phones and tablet PCs. But multitouch large displays are leading to a trend where collaborative applications are being developed for use in corporate spaces like conference and discussion rooms. Not only this, multitouch-based applications are going to penetrate into households soon," believes Tapadia.

Tapadia enumerates a few interesting applications that may shortly emerge: "Probably a digital coffee table may come up in your home, allowing users to access the Internet and use applications like Facebook, transfer images from a phone and share with friends, and so on. It will become a collaboration surface. Ten years ago, we used to play board games. Today, these games have been

replaced with playstations and mobile-based games. With touch surfaces penetrating into households, the era of boardgames might come back and we may be soon playing snakes and ladder on a digital table."

While we may shortly see lots of apps being developed for gaming, household use, retail and corporate spaces, mainly for large-format displays, it is important for a content developer to know how many people would be interacting with the surface at the same time, and what kind of content would be required to engage them with the multitouch capabilities.

Touch and gesture combo. Engagement of customers is essential and anything that makes the interaction more natural or exciting, helps. No wonder gesture-based interaction is becoming more commonplace.

"With the release of a consumer 3D depth camera on the Xbox and soon PC, users are looking at having their touchscreens to be controlled 'off-screen,' i.e., from a slight distance. Overall, 3D touch, flexible touch panels, and the convergence of touch, gesture, motion and voice are going to be hot areas in 2012 and beyond," says Sewell. In the distant future, in-cell or on-cell technologies will likely emerge as the technologies of choice.

What's in the offing?

A lot of research is happening to make touchscreen technology more commonplace and improve the quality of the experience rendered.

Sewell reveals, "Touch International is working on a few techniques to enhance the quality of touch displays so that one cannot see the traditional diamond pattern that many touchscreens have, giving users incredible optics."

"Apart from this, we have come out with projected-capacitive touch screens that are also able to work under a water spray. This allows our products to be used in marine and outdoor applications that were traditionally off-limits to touch. Other breakthroughs include non-burning, self-extinguishing, low-toxicity touch panels for in-flight entertainment, and

touch panels with EMI shielding to satisfy even the strictest interference requirements. For projected-capacitive, we are currently working to scale its multi-touch sensors up to 58.4 cm (23 inches), providing the option of thicker cover lenses and enhancing its proximity-sensing capabilities. We are also developing ultra-thin PCB electronics that utilises heat stamping to eliminate the need for connectors, serving the same purpose as a chip-on-flex controller at a fraction of the price. Work is also underway to develop a sunlight-readable resistive screen which would provide industrial and military customers with an enhanced outdoor solution."

GestureTek has developed CUBE—an innovative plug-and-play solution for interactive floors and walls which does not require a technician for installation. After removal from the box it runs, as it is, with a 229-279cm diagonal screen size on the floor, wall or counter. "Soon new 3D depth cameras with a higher resolution will hit the market, and that will allow really sophisticated track and gesture control matrices to emerge," opines Vincent.

Tapadia divulges a few details about the work that his firm is doing to come out with cost-effective materials that are likely to bring down the cost of touch-based solutions: "We are trying to reduce the number of components that go into the manufacture of the touchscreen. By the use of infrared optical sensors, we are trying to get a lower-cost screen at a lower packed density of electronics on the circuits. This will help bring the component and manufacturing cost down."

In a nutshell

Consumers are accepting touch as one of the most natural ways of interaction with the digital world. It has become imperative for design engineers to include touchscreen technology in their repertoire of tools and make the most of it by scientifically selecting the best option for their project from the myriad of choices available in this domain. ■

The author is an executive editor at EYF